PAYTM BUILD FOR INDIA AI HACKATHON — PUNE · NOV 15, 2026

RTO Trust Layer

The merchant's AI risk teammate for Cash-on-Delivery — bounded, explainable, and regulator-ready by construction.

ACCEPT | REVIEW · OTP | REJECT

TEAM

Team RTO Trust Layer

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Track 1 · Merchant Growth AI

PROOF-OF-LIFE

github.com/Neeraj-Parekh/special-parakeet

rto-trust-layer.vercel.app

397 tests passing · 22 endpoints live

01 · PROBLEM STATEMENT

Cash-on-delivery is India’s largest unpriced credit risk.

Every COD order extends a stranger a 3–7 day uncollateralized loan. When it comes back, the merchant pays — courier both ways, refund, refund, inventory stuck for weeks.

₹50,000 Cr

LOST TO COD RETURNS EVERY YEAR (INDUSTRY EST.)

~23%

AVERAGE RTO ACROSS 180M+ SHOPPERS (GOKWIK)

58%

FESTIVE-QUARTER COD RETURNS VS <15% PREPAID

₹200–250

NET LOSS ON A ₹1,000 RTO ORDER — ZERO REVENUE

12×

A FAILED DELIVERY COSTS VS A VERIFICATION CALL

Why incumbents don’t fix it

Pincode-level flags — block entire neighbourhoods, not risky customers

Black-box scores — merchant can't see WHY an order was flagged

No audit trail — nothing to show a regulator or a CFO

Sources verified 2026-11-06 · analysis/ECONOMICS_OF_COD_TRUST.md in the repo.

01 · PROBLEM STATEMENT — WHY NOW

Three clocks converged in 18 months.

OCT 2025

NPCI OC-201B — UPI Circle

Delegated payments got hard caps: **₹5,000** per transaction, **₹15,000** per month, 5 devices, 24h cooling. Any AI agent moving money on UPI must live inside these rails — most agent demos ignore them.

JUN 2026

RBI draft Model Risk Management guidance

Tamper-evident audit trails, human-in-the-loop overrides, red-teaming, kill switches — mandated for every AI model in Indian finance. Today's black-box RTO shields cannot comply.

2025-26

Zero-MDR economics

>96% of consumer UPI payments are under ₹2,000 — payment margin is ~zero (PCI estimates ₹5,500 Cr/yr subsidy). Platform value moved to merchant services: Paytm's own pivot — **13.7M** merchant device subscriptions, merchant lending, agentic AI.

The window: whoever builds trust infrastructure for merchant-side AI now sets the standard the regulation will demand.

02 · SOLUTION

RTO Trust Layer — a merchant risk command center with a bounded AI operator.

Not a model. Not a chatbot. A platform where every decision is **cost-optimal, explainable, and audit-sealed**.

Decide 01

Every order gets a 3-way verdict — **ACCEPT / REVIEW / REJECT** — with thresholds derived from the merchant's own cost structure (Bahnsen cost-sensitive BMR), never a static pincode blocklist.

ACCEPT | REVIEW · OTP | REJECT

Explain 02

Inline TreeSHAP ships every decision with ranked, signed reason codes:

```
"73% risk because: COD + ₹12,400 + new customer + vague address."
```

A CFO can read it. A regulator can verify it.

Guard 03

The AI operator works inside a **7-action allowlist**. Refusals are enforced by a deterministic classifier **BEFORE** any LLM call — `"block order"` can never return anything but refused. Money-moving actions need dual-control HMAC co-signs.

Live today: rto-trust-layer.vercel.app · vendor landing + merchant console + /story three-pane demo · GitHub: Neeraj-Parekh/special-parakeet

02 · SOLUTION — HOW IT WORKS

One request path, six guarantees.

01
Order in

Checkout POSTs the order: amount, category, customer, address, PIN

→

02
Memory

79-dim feature store: per-customer RTO history, recency/frequency/monetary, device + geo graph signals

→

03
Score

ONNX HistGradientBoosting champion → calibrated P(RTO) in 1.59 µs/row

→

04
Decide

Cost-optimal BMR over the merchant's FP/FN/OTP costs → **ACCEPT** / **REVIEW** / **REJECT**

→

05
Check

Rules engine + OC-201B mandate caps (₹5K/txn, ₹15K/mo, 24h cooling) — breach returns 422, proven live

→

06
Seal

SHA-256 hash-chained + Merkle-sealed audit record with RFC 6962 inclusion proofs

● GUARDRAIL RAIL

kill-switch · circuit breaker (never fail-open) · per-IP rate limiting · drift monitors (DDM + ADWIN) · replay-nonce table · RFC 6749 token rotation

02 · SOLUTION — EVIDENCE

Validated on real data. Audited line by line.

32×

baseline lift

PR-AUC 0.3950 on Olist Brazilian e-commerce (19,784 orders, real repeat-customer history). **Baseline 0.0123.** The signal is per-customer memory — exactly what incumbents don't have.

6.05×

Amazon India champion: PR-AUC 0.1027 vs 0.017 baseline — honest about the ceiling: no user history in that dataset (ceiling ≈0.12).

34%

of loss prevented

Selective OTP verification: challenge only orders where intervention cost < expected loss (Drummond-Holte optimal sweep over 19 thresholds).

397 / 411

pytest suite: **397 passed, 0 failed, 14 skipped** (Postgres/Redis paths). Plus AST meta-regression guards against tautology tests.

33 real · 0 decorative

Independent **37-feature audit** with file:line evidence. Every claim in this deck maps to a file you can open.

Metrics: data/olist/artifacts/metrics.json · audit: AUDIT_REPORT.md in repo.

03 · TECH STACK

Two codebases, one contract — built like a payments company would.

Merchant surface

LIVE

01 **Next.js 16** · App Router · TypeScript

02 **Tailwind CSS 4** + shadcn/ui

03 **Prisma** + SQLite (Neon-ready)

04 **JWT auth** — RFC 6749 rotation, per-route RBAC scopes

05 **Deployed:** Vercel (git auto-deploy)

Risk engine

LOCAL / CLOUD

01 **Python 3.12** · FastAPI — 5,107 lines, 22 endpoints

02 **ONNX Runtime** inference · scikit-learn pipelines

03 **SHAP TreeSHAP** explainability (cached explainer)

04 **PostgreSQL** + 12 alembic migrations · Redis Streams with Kafka seam

05 **Prometheus** metrics · OpenTelemetry tracing

Operations & delivery

K8s multi-AZ manifests — topologySpread, PDB, Little's-Law-derived HPA (λ=1000 rps, p99 40ms ⇒ L=40)

Terraform (multi-AZ RDS + MSK + EKS), **Docker Compose 9-service stack**, **7-stage TFX MLOps pipeline** with PR-AUC ≥ 0.60 deploy gate

CI: Semgrep (owasp-top-ten), TruffleHog, Dependabot auto-merge, nightly CVE scan

04 · USP

Four things nobody else ships together.

01 Memory beats pincode

Incumbents flag postal codes. We underwrite the customer: **79-dim per-customer history** is the feature family that produced the **32× lift**. A good repeat customer in a bad PIN ships normally.

02 Bounded by construction, not by prompt

The operator console's refusals are enforced by a **deterministic classifier** that runs **BEFORE the LLM**. Read `src/app/api/copilot/route.ts`: there is no code path where "block order" returns anything but refused. OC-201B caps are enforced the same way — **422 on breach**.

03 Regulator-ready by design

Merkle-sealed audit (RFC 6962 proofs), **dual-control HMAC overrides**, kill-switch, drift monitors, red-team docs — mapped 1:1 to RBI's June 2026 MRM guidance in `docs/RBI_MRM_MAPPING.md`.

04 The merchant owns the economics

Rules DSL (compile-checked, no eval) + **live cost-curve sliders**. Thresholds come from the merchant's own FP/FN costs — not a black box they can't tune.

Incumbent comparison kept honest: incumbents have distribution (millions of merchants); we have the decision infrastructure distribution needs.

05 · IMPACT & BENEFITS

Everyone in the COD chain gets paid back.

Merchant

- **₹200–250 saved** per prevented ₹1,000 RTO order.
- Inventory cash unlocked from **60–180 day** return cycles.
- Fewer wrongful rejections = **retained revenue**.
- Selective OTP cuts verification spend by challenging only marginal orders.

Shopper

- Fewer wrongful **COD refusals** at checkout.
- OTP friction only when risk actually justifies it.
- A memory of good behaviour — repeat customers get **instant acceptance**.

Platform — Paytm

- Merchant risk services on zero-MDR rails: the monetization surface Paytm's own pivot points at (**13.7M device subscriptions**, merchant lending).
- Risk decisions are the **highest-frequency, highest-value** merchant touchpoint.
- Agentic-commerce guardrails shipped **before regulators mandate them**.

Regulator & economy

- Tamper-evident audit trails align with **RBI MRM guidance** before it becomes law.
- Formalizes **₹50,000 Cr** of informal credit.
- Kill-switch + dual-control = **auditability an inspector can verify**.

06 · BUSINESS MODEL

Priced where the value lands: per decision, not per seat.

PRICING

₹0.25

per scored order (volume tiers ₹0.10–0.50)

SaaS tier: ₹999/mo per merchant incl. console + audit exports

Pays for itself at 5 prevented RTO orders/month.

WEDGE — DISTRIBUTION

Paytm merchant base:

13.7M

device subscribers already paying monthly for merchant tooling

The risk console slots into the same subscription motion as **Paytm's Soundbox** — a decision layer on rails merchants already trust.

EXPANSION

1. **Merchant lending risk** — RTO history is a repayment signal (underwrite the same memory twice)
2. **Agentic-commerce guardrails** — license the bounded-agent + mandate-cap layer to any agent spending on UPI (OC-201B-native)
3. **RTO insurance pricing** — per-merchant loss curves make return insurance underwritable

Zero-MDR made payment margin ~zero. Merchant outcomes are the margin. We price the outcome.

07 · BUILD WITH US

Give every merchant an AI teammate they can trust with money.

Bounded agents. Explainable decisions. Audit trails regulators can verify. Built for UPI's next decade.

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